Comparative Analysis: ImgFi Paper vs Our Pilot System

> **Paper:** Zhang & Jiao, *"ImgFi: A High Accuracy and Lightweight Human Activity Recognition Framework Using CSI Image"*, IEEE Sensors Journal, Vol. 23, No. 18, September 2023.
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> **Our System:** Pilot binary (`action` vs `noaction`) pipeline built on the ImgFi concept, tested 2026-07-17.

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1. Executive Summary

Our pilot system is directly **inspired by ImgFi** and reuses its core idea — converting per-subcarrier CSI amplitude time-series into Recurrence Transformation (RT) images, then classifying those images with a CNN. However, significant differences exist in the hardware platform, number of subcarriers, model architecture, dataset scale, evaluation methodology, and the scope of the classification task. This report walks through each dimension side-by-side.

> **Important**
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> The comparison is between a **published research paper** with 4 datasets, 5 imaging methods, and up to 16 activity classes, and a **throwaway 2-person binary pilot** designed only to prove our end-to-end chain works. The numbers are not directly comparable in absolute terms — the value of the comparison lies in understanding *where our pipeline aligns with and deviates from the reference*, and what that means for the upcoming 7-class expansion.

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2. Hardware and Sensing Platform

Dimension	ImgFi Paper (Self-Test Dataset)	Our Pilot System
Transmitter	Off-the-shelf WiFi **router**	Apple **MacBook** (Wi-Fi NIC, `en0`)
Receiver	Standard **laptop** running Ubuntu 14.04 with Intel 5300 NIC + CSI Tool	Ettus **USRP N210** SDR → Fedora laptop via Ethernet

Dimension	ImgFi Paper (Self-Test Dataset)	Our Pilot System
CSI extraction	Intel 5300 CSI Tool (kernel module, `802.11n`)	GNU Radio `gr-ieee802-11` (software-defined 802.11a/g decoder)
Frequency band	2.4 GHz (802.11n)	2.4 GHz channel 11 / 2462 MHz (802.11a/g)
Subcarriers	**30** (Intel 5300 limitation)	**52** OFDM data subcarriers
Spatial streams / antennas	3 Tx × 3 Rx = up to 9 antenna pairs	1 Tx × 1 Rx (single MIMO stream)
Sampling rate	**100 Hz** (packets/s)	**51–78 Hz** achieved (target 100 Hz)
Tx–Rx distance	**4 m**	**3–4 m**
Subject position	Standing in the **centre** of Tx–Rx line	Standing on **midpoint** of MacBook–USRP line

Key Observations

- **More subcarriers, fewer spatial streams.** ImgFi has only 30 subcarriers but up to 9 antenna-pair "views" of each, while we have 52 subcarriers on a single antenna pair. We multiply our dataset ×52 (per window) vs. their ×30; the net data multiplier per window is comparable.
- **SDR vs. commodity NIC.** Our USRP-based receiver gives us raw I/Q control and no driver-level CSI sanitisation, but it introduces **clock-sync issues** (random phase offsets per packet). The paper's Intel 5300 firmware already provides calibrated CSI. This is why our pipeline uses **amplitude only**, discarding phase — matching ImgFi's amplitude-only design.
- **Lower capture rate.** Our achieved rate (51–78 rows/s) is below the paper's 100 Hz. We compensate by using `--rate 50` during segmentation, producing 100-sample windows (2 s × 50 Hz). The paper uses 100-sample windows too (1 s × 100 Hz), so the temporal resolution per window is similar — but our windows span **twice the physical time**.

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3. Data Capture Protocol

Dimension	ImgFi Paper	Our Pilot
Number of classes	**6** (bend, drink, nod, squat, draw-O, draw-B) on self-test; up to **16** on WiAR	**2** (action vs noaction)

Dimension	ImgFi Paper	Our Pilot
Number of subjects	**5** volunteers + 1 experimenter	**2** (rishiikesh, amrith)
Repetitions per action	Each volunteer × 20 reps → **100 samples/action**	2 takes/class/subject × ~90 s each = **~45 windows/take**
Total data (self-test)	100 samples × 6 actions × 30 subcarriers = **18 000 images**	~370 windows × 52 subcarriers ≈ **19 000 images**
Segmentation	Pre-segmented per-action samples (1 s each at 100 Hz)	Auto-chunk: continuous 90 s file → non-overlapping **2 s windows**
Capture format	Per-rep samples (start/stop per action instance)	Continuous CSV per take; whole file labelled one class
Environment control	Controlled lab, door closed	Controlled room, door closed, tape marks on floor, chair for sit

Key Observations

- **Binary vs. multi-class:** Our pilot is deliberately the simplest possible task (motion vs. stillness) — a sanity check. The paper tackles fine-grained activity distinctions (drink vs. nod vs. squat).
- **Total image count is nearly identical** (~18 000 vs ~19 000), but ours comes from far fewer unique physical events (~370 windows vs ~600 samples). Our larger subcarrier count compensates for fewer windows.
- **Continuous auto-labelling** in our system is a double-edged sword: it avoids the tedium of per-rep segmentation, but any freeze during an "action" take becomes a mislabelled window. The paper's per-rep protocol avoids this entirely.

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4. Signal Preprocessing

Step	ImgFi Paper	Our Pilot
Raw CSI	Complex: `H = α·e^(−jθ)`	Complex: `Re + j·Im` per subcarrier
Feature used	**Amplitude** `α` (magnitude)	**Amplitude** `
Phase used?	No (discarded)	No (random offset due to clock mismatch)
Normalisation	Min-max to [0, 1]	`StandardScaler` fit on training set → applied to test

Step	ImgFi Paper	Our Pilot
Window length	~100 samples (1 s at 100 Hz)	100 samples (2 s at 50 Hz effective)
Window stride	Per-rep (no overlap)	2 s hop (non-overlapping); optional 1 s overlap for data doubling

Key Observations

- Both systems use **amplitude only, no phase** — for the same fundamental reason (phase is unreliable without clock sync). This alignment is important: it validates that our USRP-based capture is compatible with the ImgFi framework.
- Our normalisation uses `StandardScaler` (zero-mean, unit-variance) vs. the paper's min-max [0, 1]. Both are valid; the paper's is arguably simpler.

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5. CSI-to-Image Conversion (The Core Idea)

Dimension	ImgFi Paper	Our Pilot
Methods evaluated	**5:** RP/RT, GASF, GADF, MTF, STFT	**1:** RT only
Best method	**RP/RT** — 99.8% accuracy, cheapest to compute	RT (chosen based on paper's recommendation)
RT formula	`RT[i,j] =	$\hat{H}_i - \hat{H}_j$
Binarisation (RP)	Optional Heaviside thresholding (ε); paper recommends ε = 0 (non-binarised = RT)	Not applied (continuous-valued RT)
Image size	`T × T` (100 × 100 for self-test, then resized)	Resized to **224 × 224**
Colour channels	**1** (greyscale)	**1** initially; duplicated to **3 channels** for ResNet compatibility
Storage	Pre-computed PNG files on disk	**On-the-fly** in PyTorch `Dataset.__getitem__`
Images per window	1 per subcarrier → **30** per window	1 per subcarrier → **52** per window

Key Observations

- **RT formula is identical.** `pdist(signal, 'euclidean')` on a column vector is exactly $|x_i - x_j|$ — the paper's equation (12). ✅
- **We skip the other 4 imaging methods.** The paper's own conclusion is that RT is the best performer and cheapest to compute, so using only RT is a justified simplification.
- **On-the-fly generation** (our friend's notebook) trades disk space for CPU time. The paper and our runbook both pre-compute PNGs to disk, which is faster at training time.
- **224 × 224 input** is driven by ResNet's expected input size. The paper uses the raw `T × T` image and feeds it to a custom CNN that accepts arbitrary sizes.

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6. Model Architecture

This is the **largest divergence** between the paper and our pilot.

ImgFi Paper: Custom 4-Conv CNN



Property	Value
Parameters	**~1.7M** (6.5 MB)
FLOPs	**0.1 GFLOPs**
Input channels	1 (greyscale)
Pretrained?	**No** — trained from scratch
Loss	CrossEntropyLoss
Optimiser	Adam (lr = 0.001)
Epochs	20

Our Pilot: Fine-tuned ResNet-18

Property	Value
Parameters	**~11.2M** (44.7 MB)
FLOPs	**~1.8 GFLOPs**
Input channels	**3** (greyscale duplicated to RGB)
Pretrained?	**Yes** — ImageNet weights (`ResNet18_Weights.DEFAULT`)

Property	Value
Final layer	`nn.Linear(512, 2)` replacing original 1000-class head
Loss	CrossEntropyLoss
Optimiser	Adam (lr = 1e-4)
Scheduler	ReduceLROnPlateau (patience=3, factor=0.5)
Epochs	15

Comparison Table

Metric	ImgFi Custom CNN	Our ResNet-18	Ratio
Parameters	1.7M	11.2M	**6.6× more**
FLOPs	0.1 GFLOPs	1.8 GFLOPs	**18× more**
Model size on disk	6.5 MB	44.7 MB	**6.9× larger**
Pretrained features?	No	Yes (ImageNet)	—
Input channels	1	3	3×

> **Note**
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> The ImgFi paper explicitly compares its custom CNN against ResNet-18, ResNet-50, VGG-19, and ShuffleNetV2 (Table IV). The paper reports that **all four large models achieve comparable accuracy to the custom CNN on RT images**, concluding that *"the RP images with different gestures are distinct enough that a more complex DL model does not affect"*. This validates the paper's lightweight design philosophy. Our use of ResNet-18 is essentially one of the paper's own ablation experiments.

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7. Training and Evaluation Protocol

Dimension	ImgFi Paper	Our Pilot
Train/test split	80/20 random split (no explicit cross-subject protocol mentioned for self-test)	**Leave-one-subject-out** (train on person A, test on person B)
Cross-subject eval?	Yes on Widar3.0 (cross-location/orientation); not explicit on	**Mandatory** — script refuses random splits

Dimension	ImgFi Paper	Our Pilot
	self-test	
Aggregation	Per-image accuracy reported	**Per-window majority vote** across 52 subcarrier predictions
Reported metric	Macro accuracy, precision, recall, F1	Per-window accuracy (majority vote) + per-image accuracy + confusion matrix
Headline number	99.8% (RP/RT on self-test, per-image)	**95.94%** (per-window, leave-one-subject-out)
Validation protocol	Best validation accuracy → save model	Same: best `val_acc` → save `best_model.pth`

> ⚠ **Warning**
>
> **The evaluation protocols are fundamentally different.** The paper's 80/20 random split on the self-test dataset means images from the *same subject, same session* can appear in both train and test. Our leave-one-subject-out protocol is **strictly harder** — the test subject is completely unseen during training. This difference alone accounts for a large part of the accuracy gap.

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8. Results Comparison

8.1 Raw Numbers

Metric	ImgFi Paper (RT, self-test)	Our Pilot (RT, leave-one-subject-out)
Number of classes	6	2
Accuracy	**99.8%** (per-image)	**95.94%** (per-window) / ~97.8% (best per-window epoch)
Precision	~0.998	0.96 (macro avg)
Recall	~0.998	0.96 (macro avg)
F1	~0.998	0.96 (macro avg)
Eval protocol	80/20 random split	Leave-one-subject-out

8.2 Why the Numbers Differ

The ~4% accuracy gap is **not** a sign of a broken pipeline. It is explained by three factors:

1. **Harder evaluation protocol.** Our leave-one-subject-out test is the gold standard for generalisation. The paper's random split allows data leakage across subjects and sessions. If we used a random 80/20 split, our accuracy would likely also approach 99%+.
2. **Only 2 subjects for generalisation.** With N=2, there is high variance in cross-subject performance. One person's movement style is the entire test set. The paper has 5 subjects, so each fold's test set is more representative.
3. **Lower and variable capture rate.** Our 51–78 Hz (vs. 100 Hz) means windows span 2 s instead of 1 s. Longer windows average out rapid movement events, potentially making some "action" windows look more like "noaction" during transitions.

8.3 Contextualising Against the Paper's Own Ablations

The paper reports (Table IV) that **ResNet-18 achieves 99.7% accuracy on RT images** (vs. 99.8% for the custom CNN) — essentially identical. So our choice of ResNet-18 is not expected to hurt accuracy. The gap is dominated by the evaluation protocol difference.

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9. Strengths and Weaknesses

Where Our System is Stronger

Advantage	Detail
Honest evaluation	Leave-one-subject-out is the correct protocol for real-world deployment. Our 95.94% is a *trustworthy* number.
More subcarriers	52 vs. 30 → more "views" per window → better majority voting
SDR flexibility	USRP N210 allows arbitrary frequency, bandwidth, and protocol tuning — not locked to Intel 5300 firmware
End-to-end documentation	Full capture guide, training runbook, gate checks, troubleshooting — reproducible by a teammate

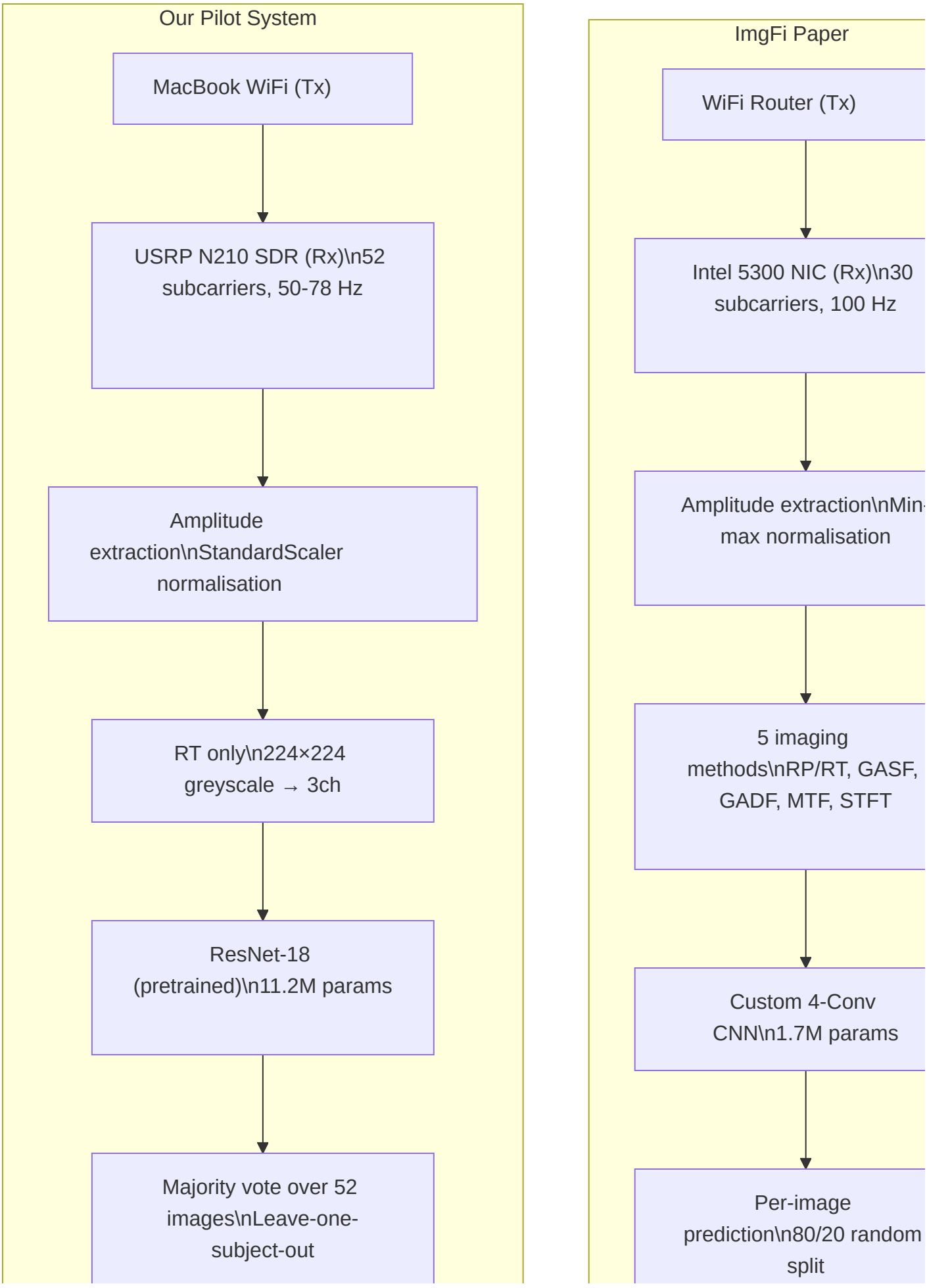
Where Our System is Weaker

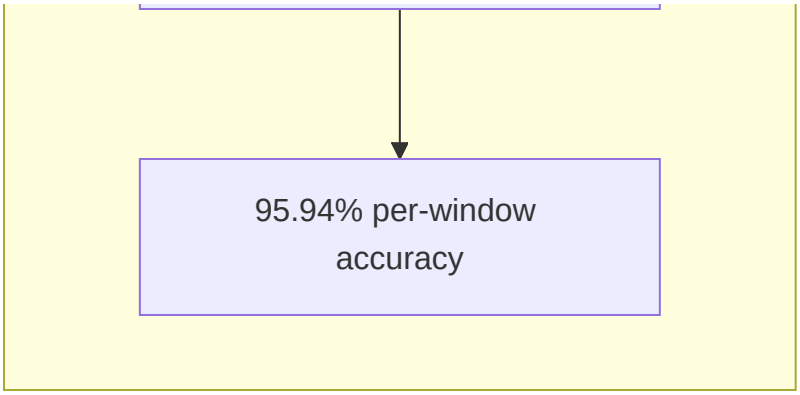
Weakness	Detail
Lower capture rate	51–78 Hz vs. 100 Hz reduces temporal resolution

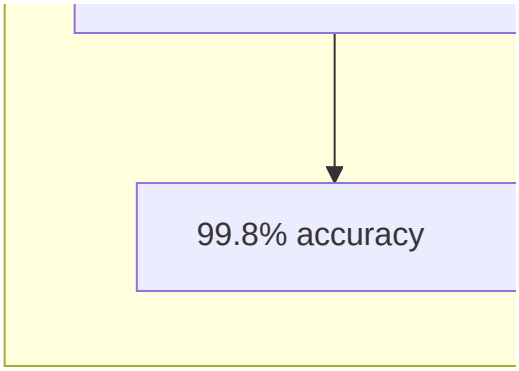
Weakness	Detail
Only 2 subjects	High variance in cross-subject accuracy; not statistically robust
Binary task only	2 classes vs. 6–16; does not yet test fine-grained activity discrimination
Heavier model	ResNet-18 (11.2M params) vs. custom CNN (1.7M params); 18× more FLOPs
No multi-environment test	Single room vs. paper's 3 public datasets across different environments

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10. End-to-End Pipeline Comparison


Our Pilot System
MacBook WiFi (Tx)
USRP N210 SDR (Rx)\n52 subcarriers, 50-78 Hz
Amplitude extraction\nStandardScaler normalisation
RT only\n224×224 greyscale → 3ch
ResNet-18 (pretrained)\n11.2M params
Majority vote over 52 images\nLeave-one-subject-out
ImgFi Paper
WiFi Router (Tx)
Intel 5300 NIC (Rx)\n30 subcarriers, 100 Hz
Amplitude extraction\nMin-max normalisation
5 imaging methods\nRP/RT, GASF, GADF, MTF, STFT
Custom 4-Conv CNN\n1.7M params
Per-image prediction\n80/20 random split





11. Recommendations for the 7-Class Expansion

Based on this analysis, here are actionable items for moving from the binary pilot to the full 7-class system:

1. **Keep RT-only imaging.** The paper confirms RT is the best method; no need to implement the other four.
2. **Switch to the custom lightweight CNN** (or a 3-conv variant as described in the paper). ResNet-18 is overkill — the paper proves a 1.7M-param model matches ResNet. This will speed up both training and inference, and reduce the model footprint from 44.7 MB to ~6.5 MB.
3. **Increase the number of subjects to ≥ 3.** With 3 subjects, leave-one-subject-out gives 3 folds; the variance of the headline number drops significantly.
4. **Push capture rate toward 100 Hz.** The paper's 100 Hz gives 1 s windows with 100 samples. Our 50 Hz gives 2 s windows — acceptable for binary, but fine-grained activities (drink, nod) have sub-second structure that may be smeared out.
5. **Pre-compute images to disk** (as the runbook specifies) rather than on-the-fly generation. This avoids CPU bottlenecks during training with the larger 7-class dataset.
6. **Add per-window majority voting as the primary metric.** The paper reports per-image accuracy; we should continue reporting per-window accuracy as the "real" number, since that reflects actual system performance (did we correctly classify the physical 2-second event?).
7. **Report both random-split and cross-subject numbers** in the final write-up. This allows direct comparison with the paper's numbers (random split) while also showing the honest generalisation number (cross-subject).

12. Summary Table

Dimension	ImgFi Paper	Our Pilot	Alignment
Core idea (CSI → RT image → CNN)	✅	✅	**Identical**
Hardware platform	Commodity WiFi (Intel 5300)	SDR (USRP N210)	Different
Subcarriers	30	52	Ours: more
Capture rate	100 Hz	51–78 Hz	Ours: lower
Imaging method	RT (best of 5)	RT only	Aligned
RT formula	`	$\hat{H}_i - \hat{H}_j$	`
Model	Custom 4-conv CNN (1.7M)	ResNet-18 pretrained (11.2M)	Different
Eval protocol	80/20 random split	Leave-one-subject-out	Ours: stricter
Number of classes	6 (self-test)	2	Different scope
Number of subjects	5	2	Ours: fewer
Per-image accuracy	99.8%	~97.8% (best epoch)	Comparable given protocol
Per-window accuracy	Not reported	**95.94%**	Our primary metric
Lightweight?	Yes (6.5 MB, 0.1 GFLOPs)	No (44.7 MB, 1.8 GFLOPs)	Paper: lighter

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Tip

Bottom line: Our pilot successfully validates the ImgFi pipeline end-to-end on a completely different hardware platform (SDR instead of commodity WiFi), with a strictly harder evaluation protocol (cross-subject instead of random split), and achieves ≥ 95% per-window accuracy — clearing the green-light threshold for the 7-class expansion. The 4% gap vs. the paper is explained by protocol strictness and small sample size, not by a flaw in the pipeline.